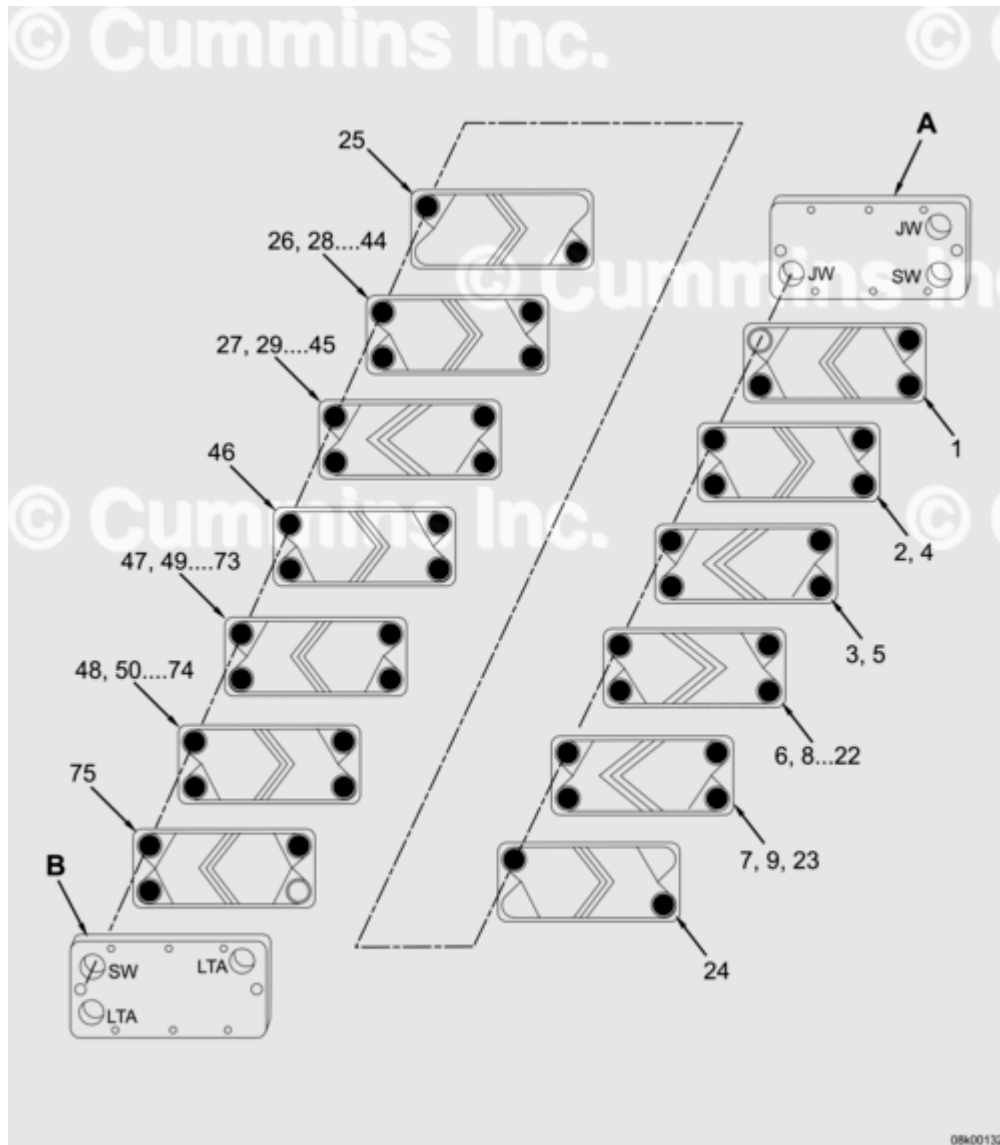


Trainings ▾

Exploded View

with Mechanically Actuated Injector



A. Pressure plate

B. Frame plate.

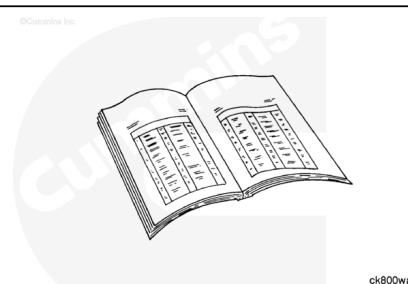
Plate Location in Heat Exchanger (See Figure 1)	Plate Identification Number (stamped in end of plate)
1	39500871-49 (A)
2, 4	374541-2403 (B)
3, 5	374241-2403 (A)

Plate Location in Heat Exchanger (See Figure 1)	Plate Identification Number (stamped in end of plate)
6,8...22	374341-2403 (B)
7, 9...23	374241-2403 (A)
24	374541-2414 (B)
25	374441-2414 (A)
26, 28...44	374541-2403 (B)
27, 29...45	374241-2403 (A)
46	374541-2403 (B)
47, 49...73	374441-2403 (A)
48, 50...74	374541-2403 (B)
75	374441-4485 (A)

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect batteries or air supply to air starter to prevent accidental engine starting.

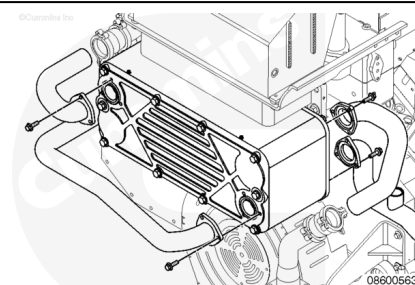
- Drain the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the expansion tank. Refer to Procedure 008-052 (Expansion Tank) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)

Remove

with Mechanically Actuated Injector

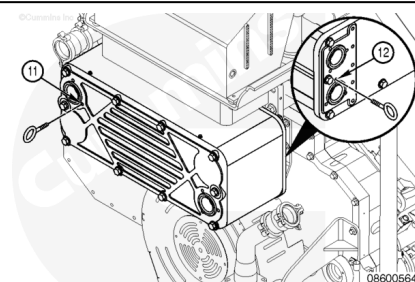
Disconnect all plumbing for the front and rear of the heat exchanger:

- Sea water tube
- Jacket water tube
- LTA tube.



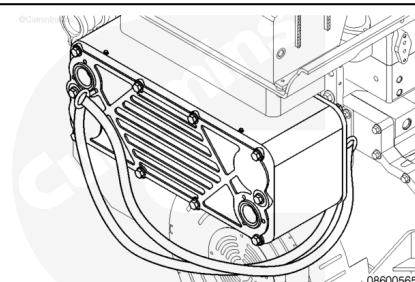
Install a lifting eye into the heat exchanger front plate at location (11).

Install a lifting eye into the heat exchanger rear plate at location (12).



The nylon strap used to remove the heat exchanger **must** have a capacity of 140 kg [300 lb].

Insert a nylon strap through the lifting eyes.

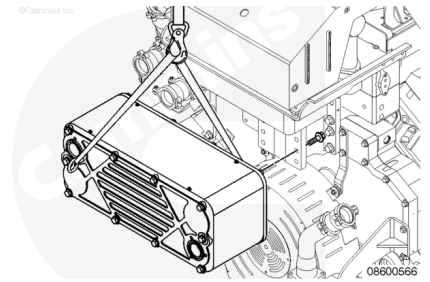


⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Remove the capscrews from the heat exchanger bracket.

Remove the heat exchanger.

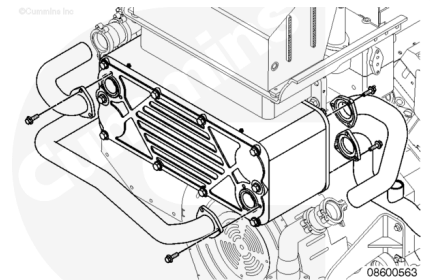


with Electronically Actuated Injector

Remove the sea water, jacket water, and low temperature aftercooling water transfer tubes.

Discard the wear sleeves and o-rings.

Remove the water transfer connections from the heat exchanger.

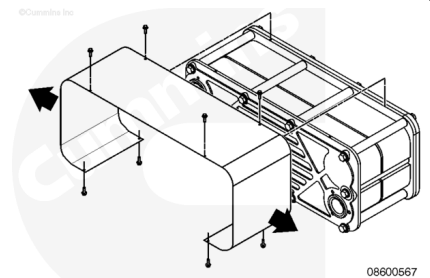


Disassemble

Remove the eight capscrews that hold the shroud to the heat exchanger assembly.

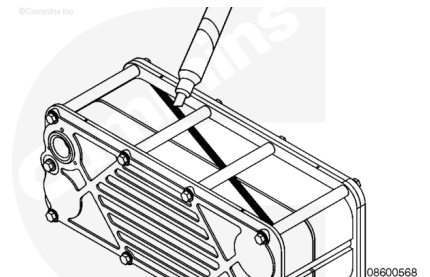
The ends of the shroud will possibly have to be bent outwards to clear it from the sides of the heat exchanger.

Remove the shroud from the heat exchanger.



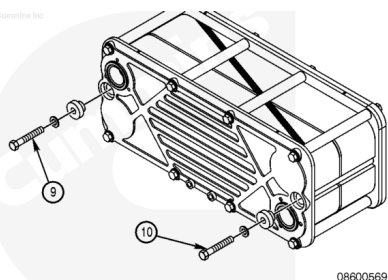
Draw a diagonal line across the top of the plate pack with a paint marker or a wide marker.

The diagonal line will aid in assembly.



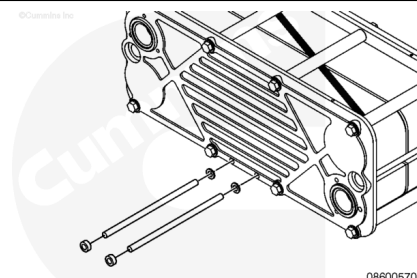
Remove spacer capscrews (9) and (10).

Remove the inserts the spacer capscrews were routed through.



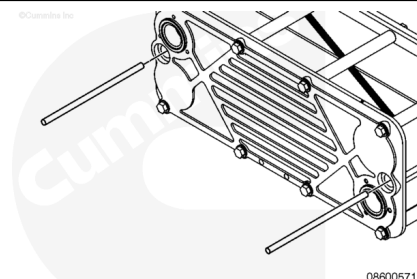
08600569

Remove the service spacers from the front cover.



08600570

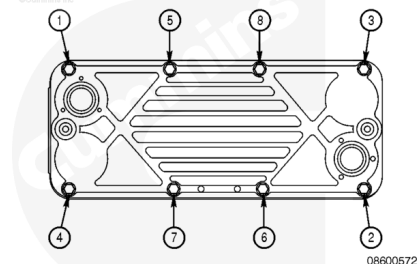
Thread the service spacers into the side alignment spacers until they bottom out.



08600571

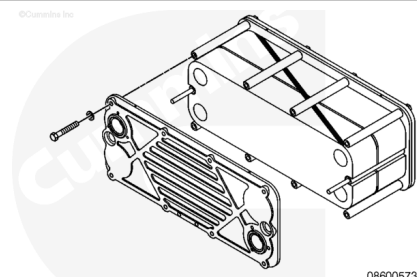
The plate pack clamping capscrews **must** be loosened in sequence and in small increments to keep the front and back covers within 12 mm [$\frac{1}{2}$ in] variance from parallel.

Loosen the plate pack clamping capscrews in sequence and in small increments until they can be removed from the assembly.



08600572

Remove the plate clamping capscrews and front cover.



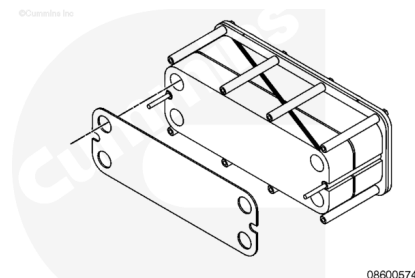
08600573

⚠ WARNING ⚠

The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

Keep the plates in the order they were removed during disassembly. This will aid in the assembly of the heat exchanger.

Remove the plates from the assembly.



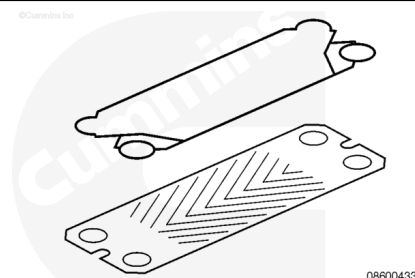
Clean and Inspect for Reuse

⚠ CAUTION ⚠

Incorrect orientation of the gaskets can cause overheating and sea water to enter the cooling system.

Observe the orientation of the gasket for assembly accuracy.

Remove each gasket from the plate one at a time.



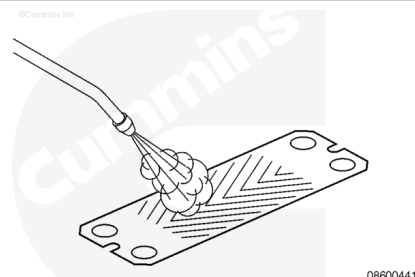
Clean the plate with a soft brush and running water.



If deposits are present that are difficult to remove, spray the plate with a high pressure water hose.

If brushing and high pressure water does **not** properly clean the plates, the deposits on the plates can be evaluated.

Use the information in the following lists to determine what solvents to use.



Incrustation

- Calcium carbonate (corrosion products - clean with nitric acid).
- Calcium sulphate (metal oxides - clean with sulfamic acid).

- Silicates (silt - clean with citric acid, alumina - clean with phosphoric acid, diatomic organisms and their excrement of various colors - clean with complexing agents EDT, A, and NTA).

Biological Growth

- Slime - Clean with alkaline cleaning agents.
- Bacteria - clean with sodium hydroxide.
- Nematodes - Clean with sodium hydroxide.
- Protozoa - Cleaning effect can be considerably increased by the addition of small quantities of hypochlorite or agents for the formation of complexes and surfactants.

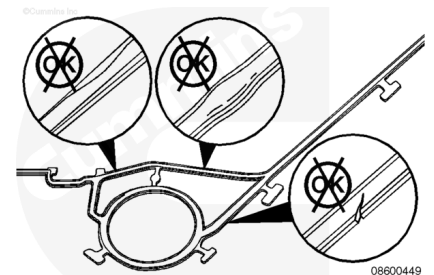
Concentration maximum: 4 Percent.

Temperature maximum: 60°C [140° F].

Solvents That Can Not be Used

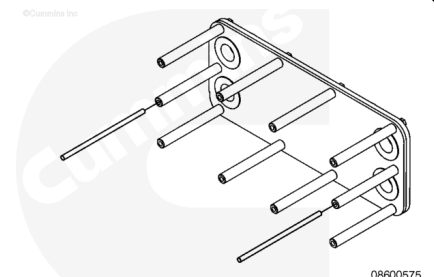
- Ketones (acetone, methylethylketone, methylisobutylketone)
- Esters (ethylacetate, butylacetate)
- Halogenated hydrocarbons (chloroethene, carbon tetrachloride, freons)
- Aromatics (benzene, toluene).

Inspect the any gaskets that are swollen, cracked, or deformed.



Assemble

Insert the service spacers into the side alignment spacers.

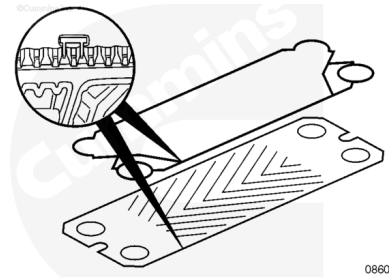


⚠ WARNING ⚠

The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

⚠ CAUTION ⚠

Incorrect orientation of the gaskets can cause over heating and sea water to enter to the cooling system.



Place the gaskets back onto the plates one at a time.

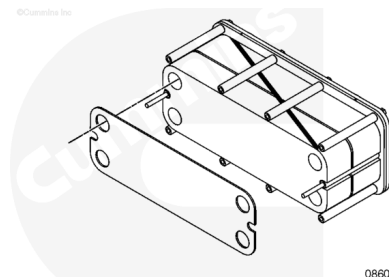
Make sure the tabs are locked into the slots and orientated correctly.

⚠ WARNING ⚠

The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

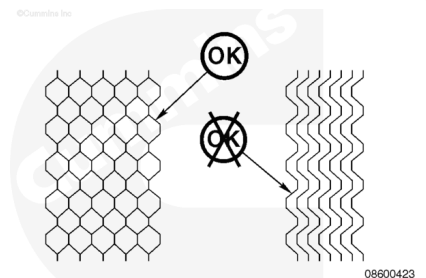
Insert the cleaned plates against the rear cover in the correct order.

Refer to the exploded view for the plate installation order.

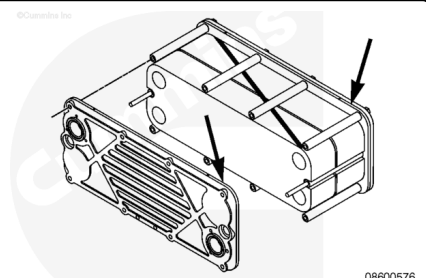


After the plates are assembled in the engine, make sure the profile shows a honey comb pattern and the gasket tabs alternate position.

If the pattern is **not** a honey comb pattern throughout the entire plate pack, the plates **must** be rearranged.



Install the front plate making sure the aligning shroud groove is on the same side (top) as the alignment groove on the rear cover.



Install capscrews (1) through (8).

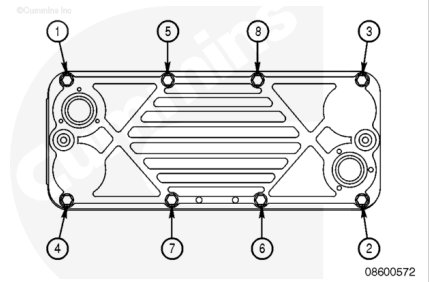
When tightening the capscrews, keep the front and back covers within 12 mm [$\frac{1}{2}$ in] variance from parallel.

Tighten the capscrews in the sequence shown in the graphic.

Torque Value: 15 n•m [133 in-lb]

Once the cover is completely against the eight spacers, tighten the capscrews in the sequence shown in the graphic.

Torque Value: 135 n•m [100 ft-lb]

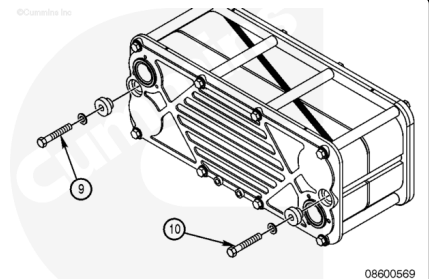


Remove the service spacers.

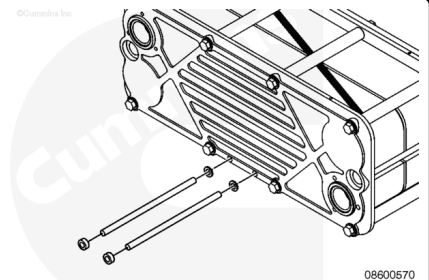
Install the inserts and the side alignment capscrews into locations (9) and (10).

Tighten the capscrews.

Torque Value: 88 n•m [65 ft-lb]



Install the service spacers into the locations shown in the graphic.



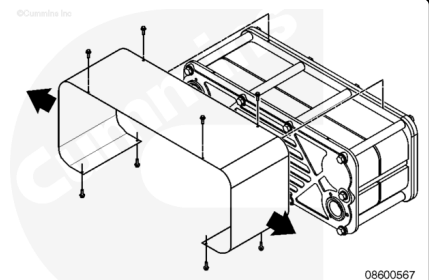
The ends of the shroud will possibly need to be bent to fit over the heat exchanger.

Install the shroud onto the heat exchanger.

Install the eight shroud capscrews.

Tighten the capscrews.

Torque Value: 9 n•m [80 in-lb]

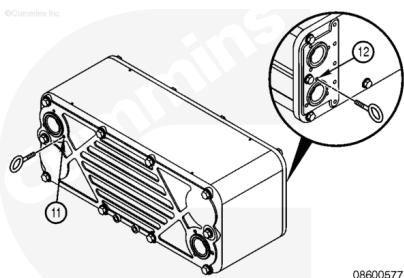


Install

with Mechanically Actuated Injector

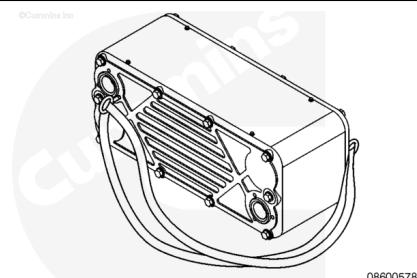
Install a lifting eye into the front plate at location (11).

Install a lifting eye into the rear plate at location (12).



The nylon strap used to install the heat exchanger **must** have a capacity of 140 kg [300 lb].

Insert a nylon strap through the lifting eyes.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

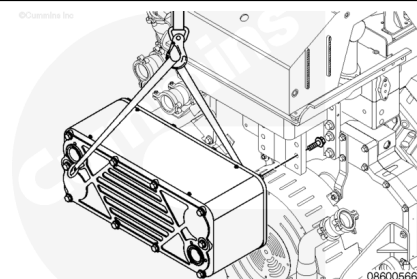
Place the heat exchanger onto the bracket.

Insert the bracket capscrews that attach the bracket to the heat exchanger.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

Remove the nylon strap.

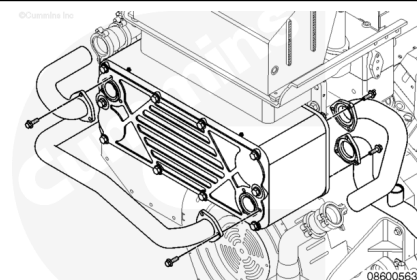


Connect all plumbing for the front and rear of the heat exchanger:

- Sea water tube
- Jacket water tube
- LTA tube.

Tighten all capscrews.

Torque Value: 45 n•m [33 ft-lb]

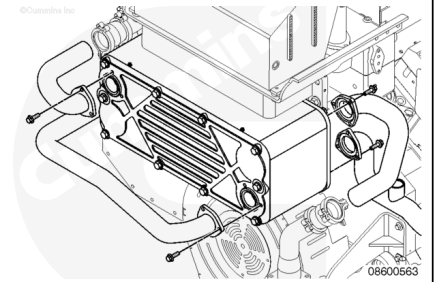


with Electronically Actuated Injector

Install sea water, jacket water, and LTA tubes on the heat exchanger.

Install new wear sleeves and o-rings.
Install castings on the heat exchanger.
Tighten all capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the expansion tank. Refer to Procedure 008-052 (Expansion Tank) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)
- Connect the batteries or air starter supply line.
- Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)